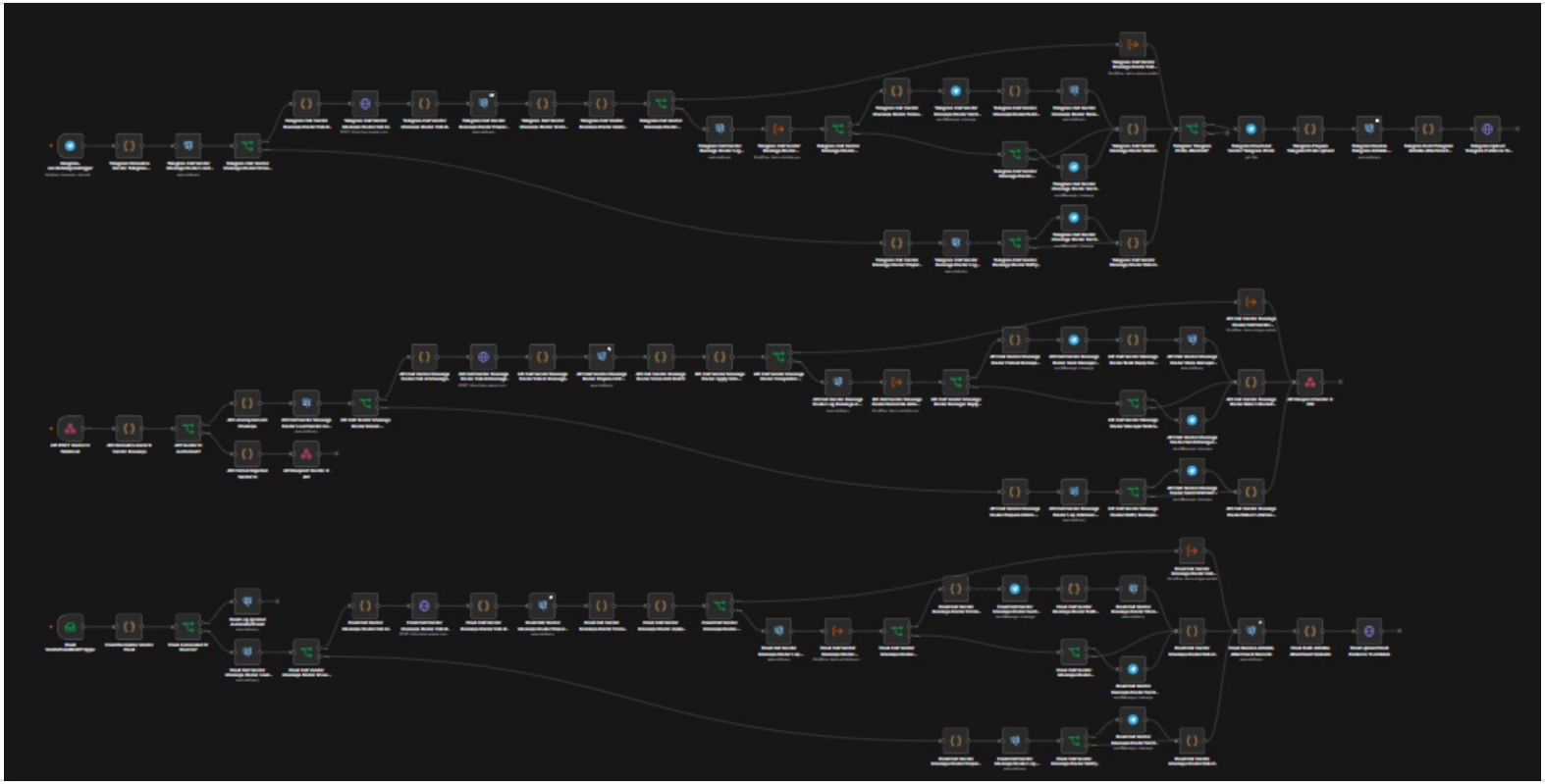
AUTOMATION CASE STUDY

Make-ready sequencing

A local-first coordination automation that turns vendor updates into guarded stage changes and manager-approved handoffs.



LIVE N8N WORKFLOW / THREE INBOUND CHANNELS

THE PROBLEM

A completed trade can sit unnoticed. Delays between paint, flooring and cleaning keep a vacant unit waiting.

THE BUILD

Seven connected workflows read updates, enforce sequence, prepare the next message and ask a manager to release it.

THE PROOF

Synthetic fixture runs tested normal updates, ambiguity, duplicates, blocked ordering and manager decisions.

SYSTEM DESIGN

One connected automation

n8n runs the steps; Postgres holds the authoritative turn and delivery state. Airtable mirrors that state for managers.

01

INBOUND

Telegram, email and a generic API normalize vendor replies.

→

02

INTERPRET

The model reads wording; fixed rules identify sender and unit.

→

03

GUARD

One shared sequence check decides whether a stage can close.

→

04

APPROVE

The manager chooses Email, Message or Both before any send.

DURABLE STATE AND OUTPUTS

POSTGRES

Turns, stages, events, approvals and channel-specific delivery records.

AIRTABLE

Manager workspace and new-turn intake, reconciled from Postgres.

TELEGRAM

Manager cards, controls, alerts and digests; vendor demo messages.

EMAIL

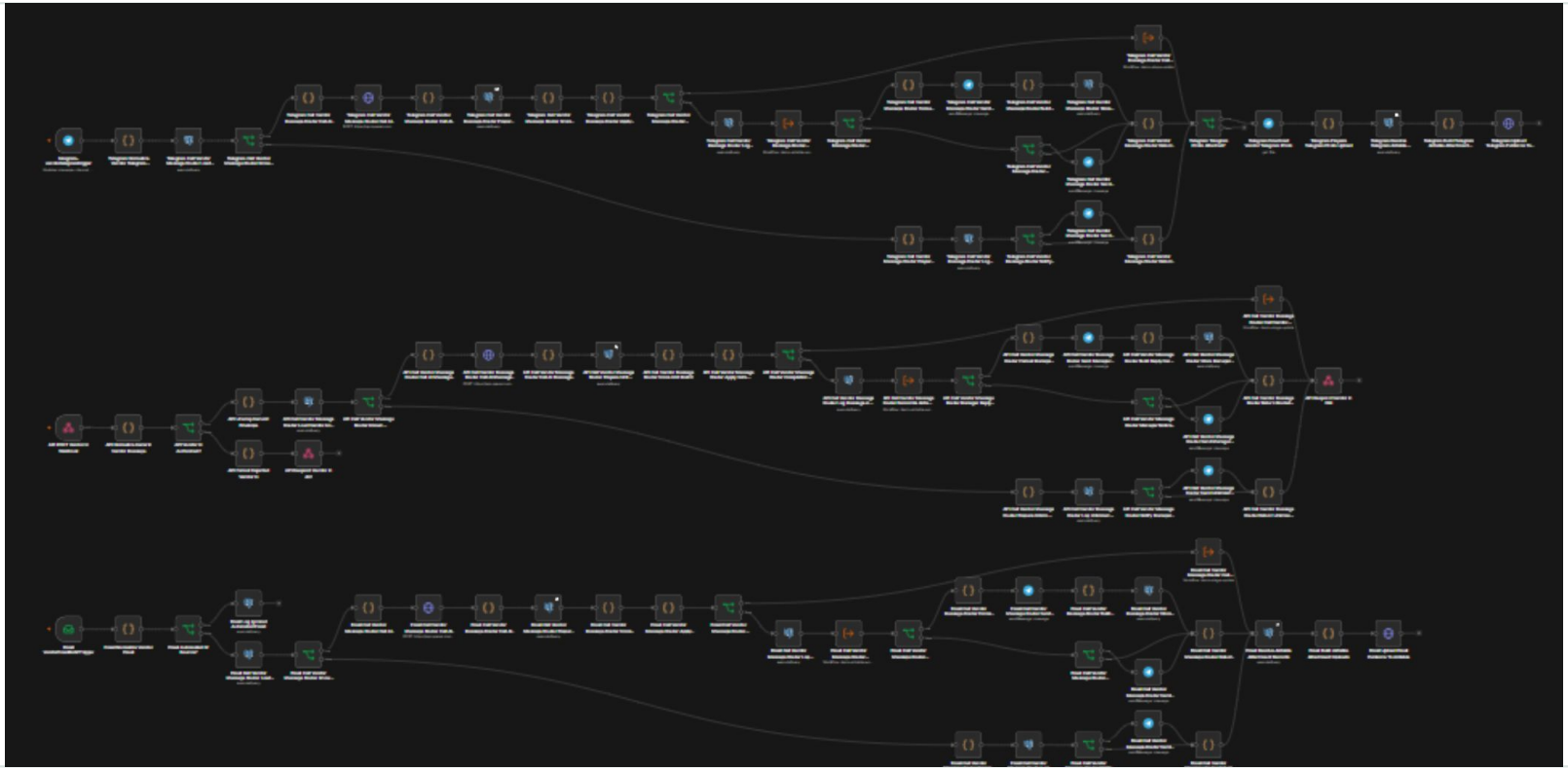
Vendor replies in; approved messages out through the mailbox.

HUMAN RELEASE POINT

The delivery workflow has no independent trigger. It is called only after a manager selects a channel on a Telegram approval card.

Vendor Communications

Three entry routes share the same interpretation and safety policy.



TRIGGER

A Telegram post, a mailbox reply or a POST to the generic vendor endpoint.

KEY STEPS

Normalize text and media; resolve the sender and conversation; interpret intent.

DECISION

Match only assigned units. Ambiguous, unsafe or duplicate messages go to review.

OUTPUT

A safe completion request, a manager notice, or a logged no-change outcome.

Three entry paths

Each row below is a direct crop of the live n8n canvas.

TELEGRAM / NORMALIZE AND ROUTE



Incoming chat updates and photos enter the same vendor policy used by the other channels.

TELEGRAM / MATCH, REVIEW, COMPLETE



The assigned-vendor match and safe-message policy determine whether Stage Update is called.

Email and thread context

A reply can resolve through its message thread before unit text is considered.

EMAIL / READ AND IDENTIFY



Mailbox headers and sender identity are normalized before interpretation.

EMAIL / DECIDE AND LOG



Questions and unclear matches go to the manager; a safe completion reaches the shared handler.

Generic inbound API

The API route proves that the sequencing logic is not tied to one messaging tool.

API / AUTHENTICATE AND NORMALIZE



The webhook checks its demo token and creates the same message envelope.

API / APPLY THE COMMON POLICY



The same sender, unit and intent decisions are applied before any state change.

Stage Update

This shared handler is the gate for vendor reports and manager Done controls.



TRIGGER

Called by a safe vendor completion or a manager completion action.

KEY STEPS

Reload current stage state and evaluate fixed prerequisites.

DECISION

Block an out-of-order or repeated completion; allow a recorded override.

OUTPUT

Close the stage, prepare a factual next-vendor draft and show an approval card.

Guard, completion and draft

The model may polish draft wording; it never chooses the unit or bypasses ordering.

LEFT / ONE SHARED ORDER CHECK



The guard checks prerequisites such as Repairs before Paint and Paint before Flooring.

RIGHT / DRAFT AND APPROVAL

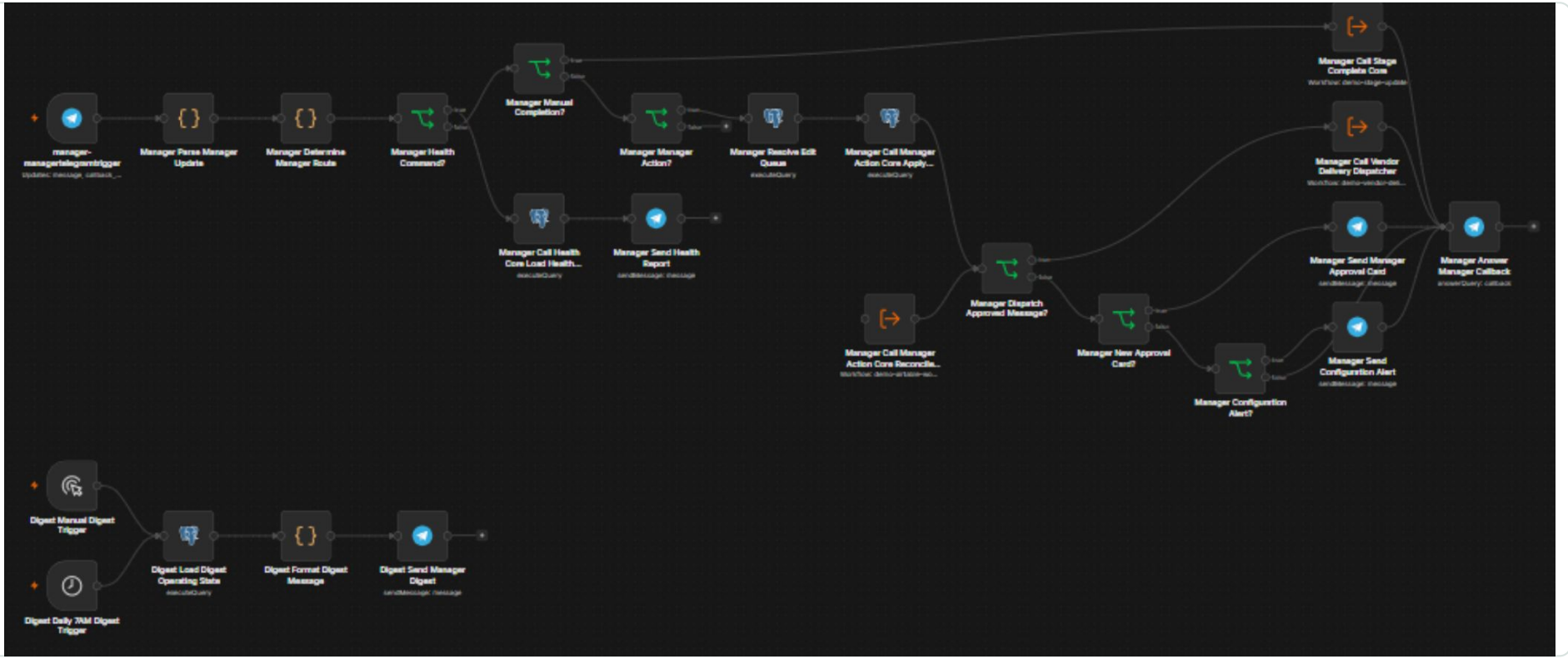


A saved draft is sent to the manager for approval. No vendor receives it yet.

WORKFLOW / MANAGER-OPERATIONS / 24 NODES

Manager Operations

Telegram is the control surface for approvals, exceptions, manual updates and the daily digest.



TRIGGER

Manager bot message or callback, plus the morning digest schedule.

KEY STEPS

Parse the requested action, load context and show the relevant control path.

DECISION

A manual Done uses Stage Update; a send choice calls Vendor Delivery.

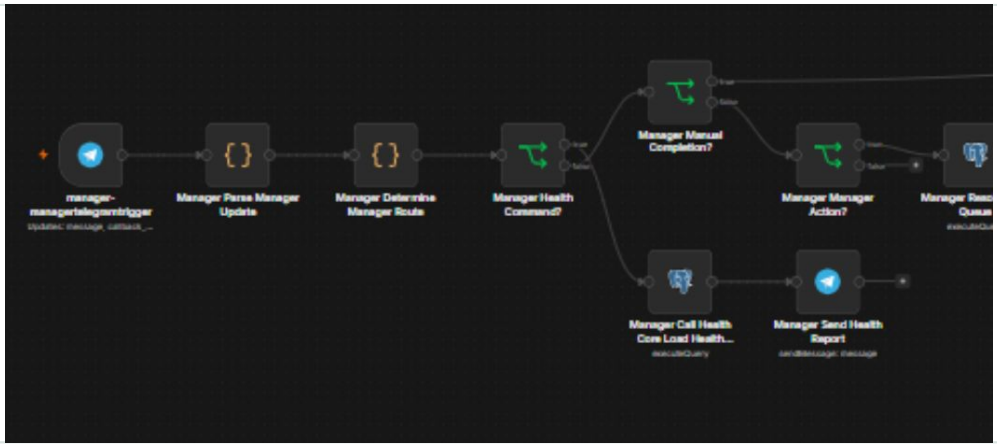
OUTPUT

A logged manager action, callback answer, digest or approval result.

Manager callback path

The control path runs left to right; every branch returns a logged outcome.

CONTROL PATH / ENTRY AND ROUTING



A manager callback is parsed into Done, approval, edit, skip, snooze, chase or cancel.

CONTROL PATH / APPROVAL AND RESPONSE

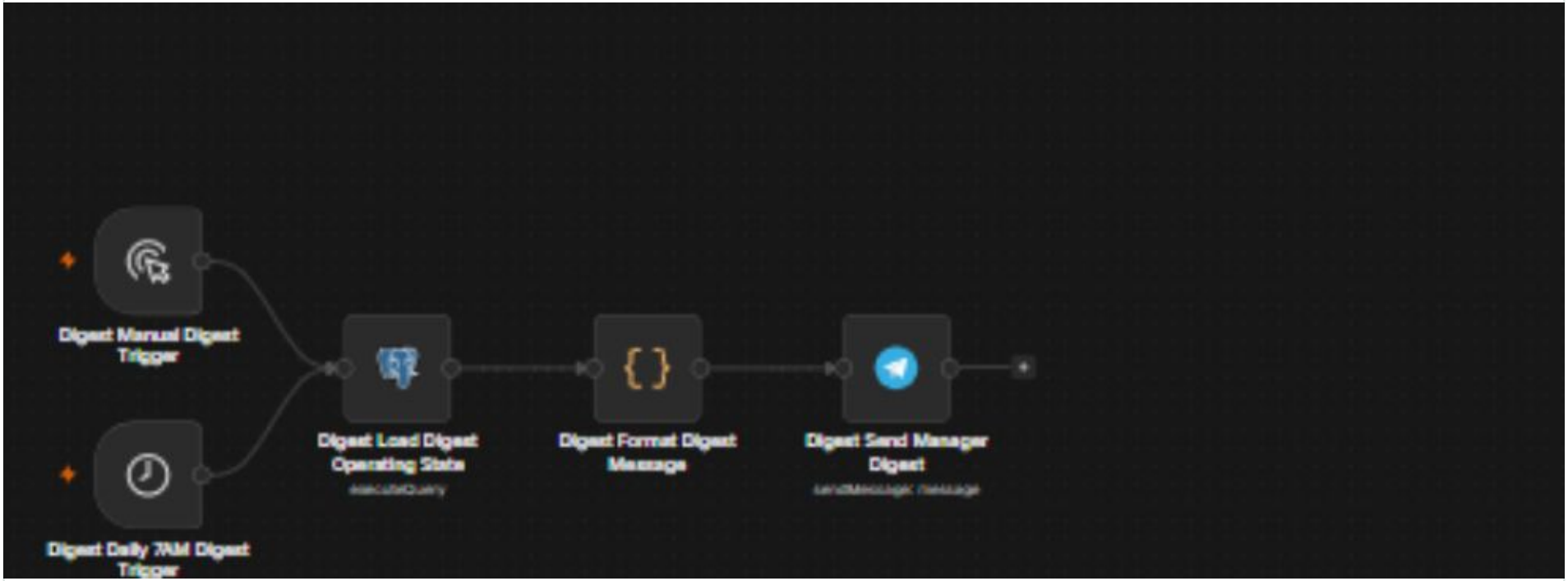


Only a channel choice dispatches Vendor Delivery. The bot then acknowledges the action.

Daily operating digest

The second branch assembles a worst-first summary with manager controls.

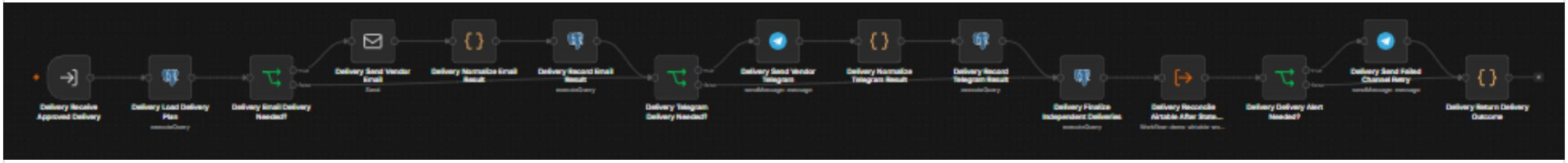
DIGEST / LOAD, FORMAT AND SEND



The scheduled run loads stage ages from Postgres, formats the worst-first summary and sends manager controls in Telegram.

Vendor Delivery

One approved queue item becomes one attempt per authorized channel.



TRIGGER

Called only from a manager-approved channel choice.

KEY STEPS

Load the destination and send Email, Message or both as selected.

DECISION

Record each provider result separately; retry only a failed channel.

OUTPUT

Delivery rows, queue outcome and next-stage activation after one success.

Independent channel records

The two send branches meet at a database finalization step.

LEFT / EMAIL ATTEMPT



Email is attempted only when the manager authorized it and a destination exists.

RIGHT / MESSAGE, FINALIZE, ALERT

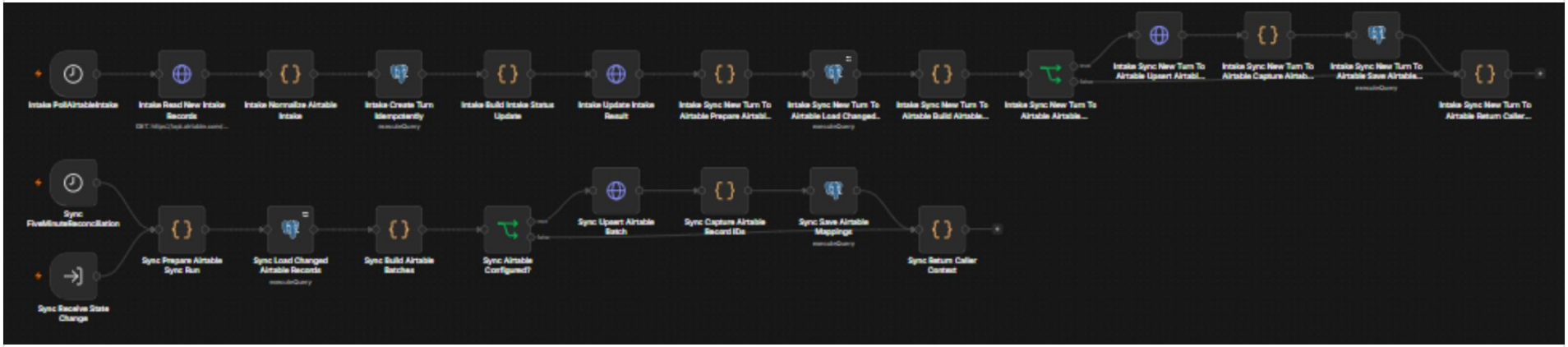


Telegram delivery is independent; the manager sees a blocked or failed route.

WORKFLOW / AIRTABLE-WORKSPACE / 24 NODES

Airtable Workspace

A manager creates a turn in Airtable; reconciliation mirrors durable Postgres state back to Airtable.



TRIGGER

A new intake record, scheduled reconciliation or a state-change call.

KEY STEPS

Validate intake, create the turn, then batch changed records for sync.

DECISION

Idempotent mappings and payload hashes prevent duplicate mirror rows.

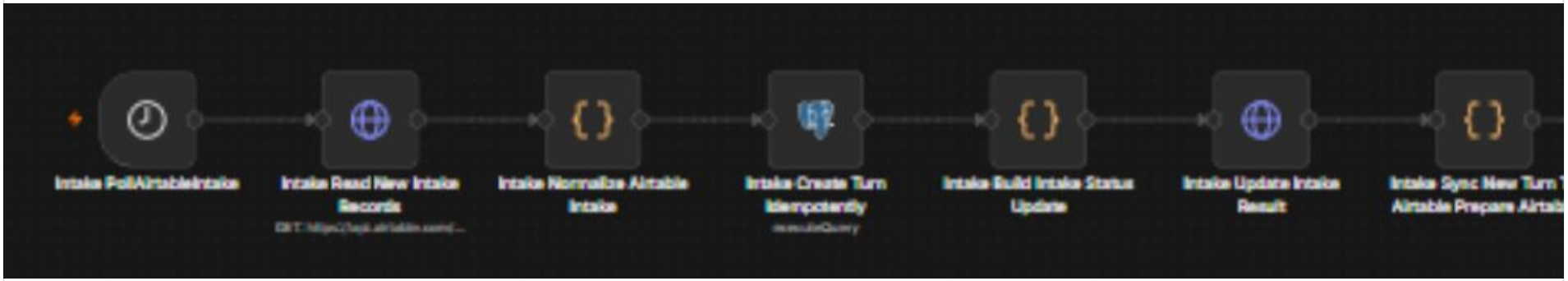
OUTPUT

Updated Turns, Stages, Vendors, Communications and approval views.

New-turn intake

The upper lane creates one turn from a manager form submission.

INTAKE / READ AND CREATE



The poll finds unprocessed intake records and creates Postgres turn stages idempotently.

INTAKE / REPORT RESULT



The intake record is updated with its result while the normal sync path starts.

State reconciliation

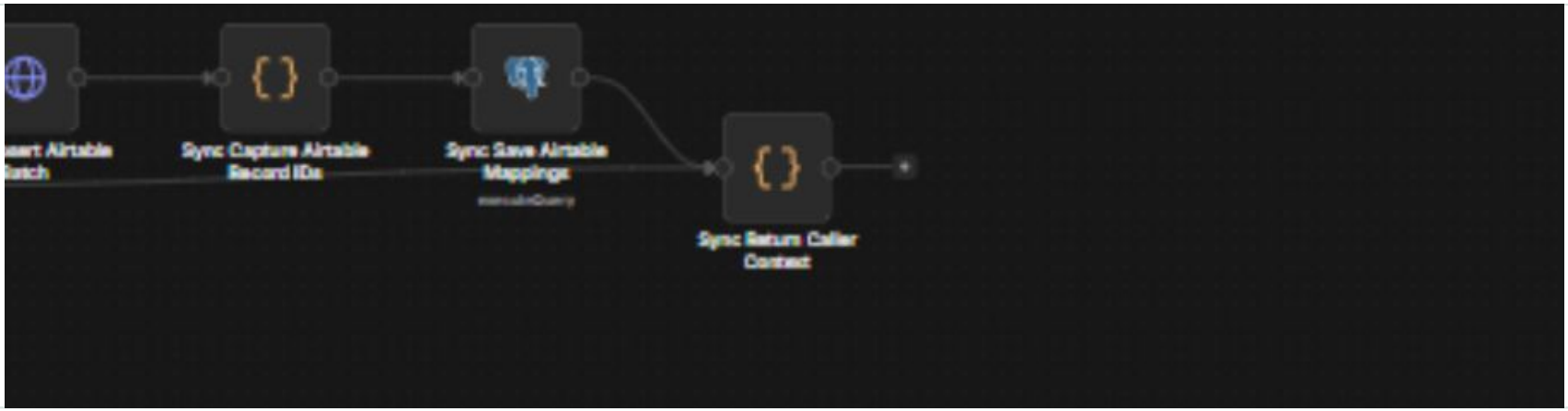
The lower lane mirrors changed records without changing the authoritative sequence.

SYNC / LOAD CHANGES



A timer or state-change call loads records with changed payload hashes.

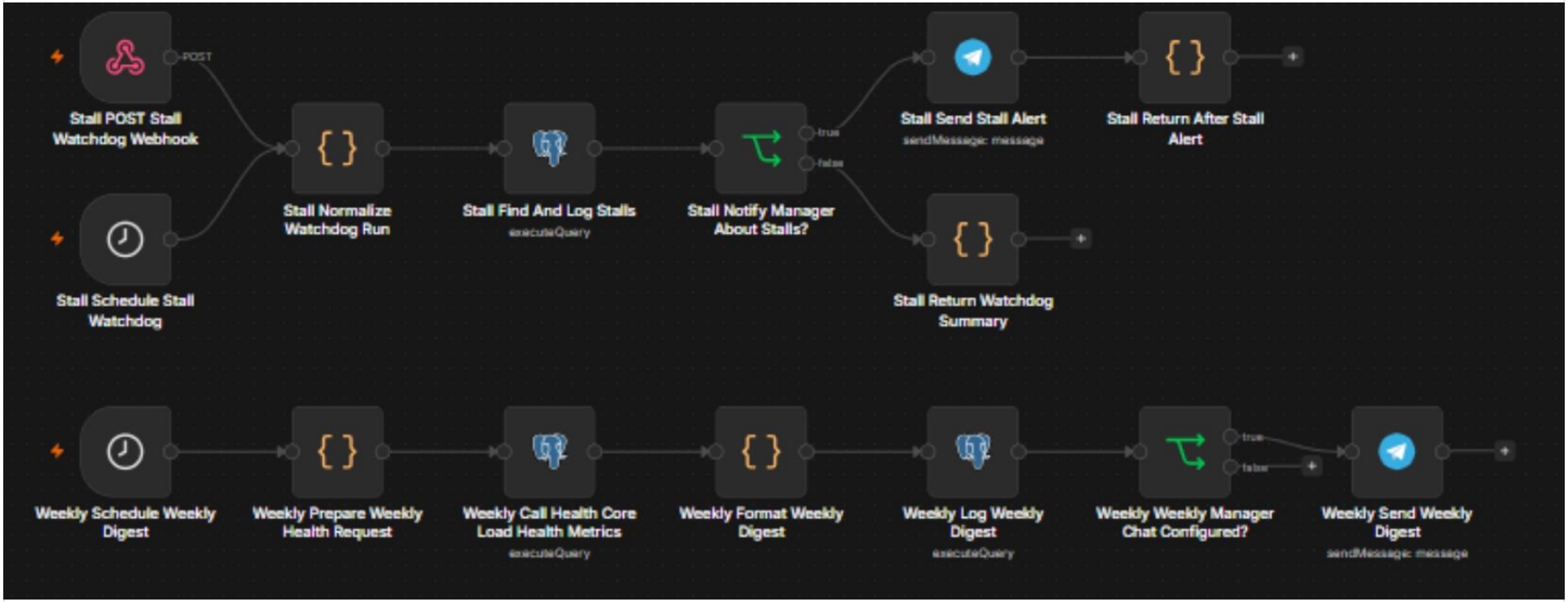
SYNC / BATCH AND MAP



Batches update Airtable, then external record IDs are retained for the next pass.

Monitoring

Silence becomes a manager-visible event instead of an invisible delay.



TRIGGER

A watchdog schedule or webhook; a separate weekly digest schedule.

KEY STEPS

Compare active stages with SLA age and prior alerts.

DECISION

Dedupe, snooze and business-hour rules decide when to alert.

OUTPUT

Manager stall alert and weekly health summary, with logged events.

Stall watchdog

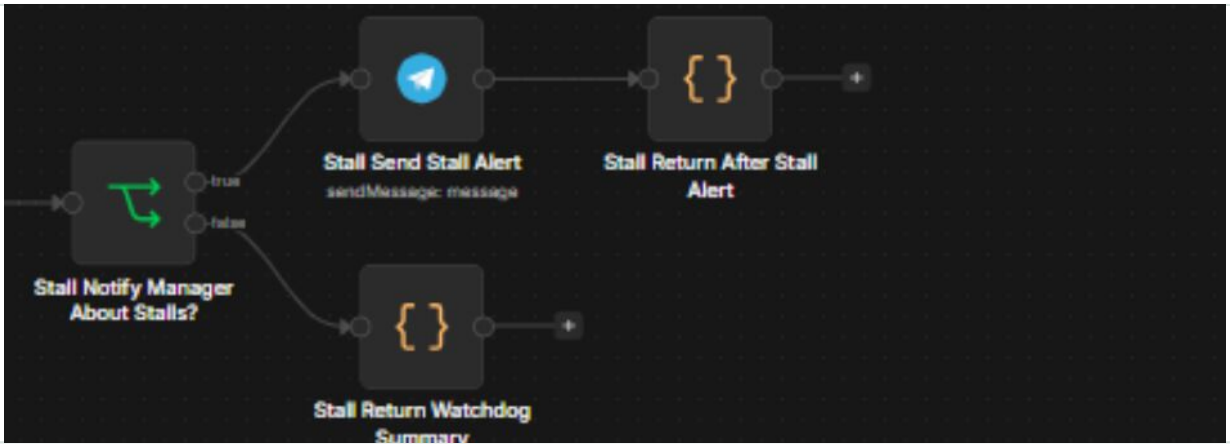
The upper lane finds overdue work and sends one actionable manager alert.

WATCHDOG / FIND AND LOG



A database query checks age, snooze state and whether this stage was already alerted.

WATCHDOG / NOTIFY

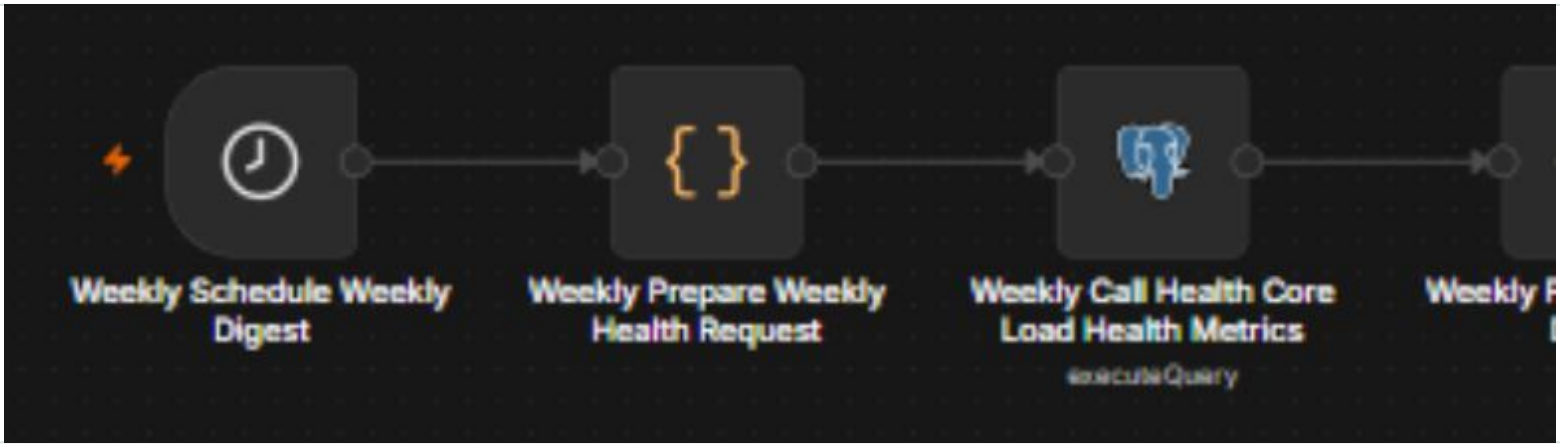


Only an eligible stall reaches the manager bot; otherwise the run exits quietly.

Weekly digest

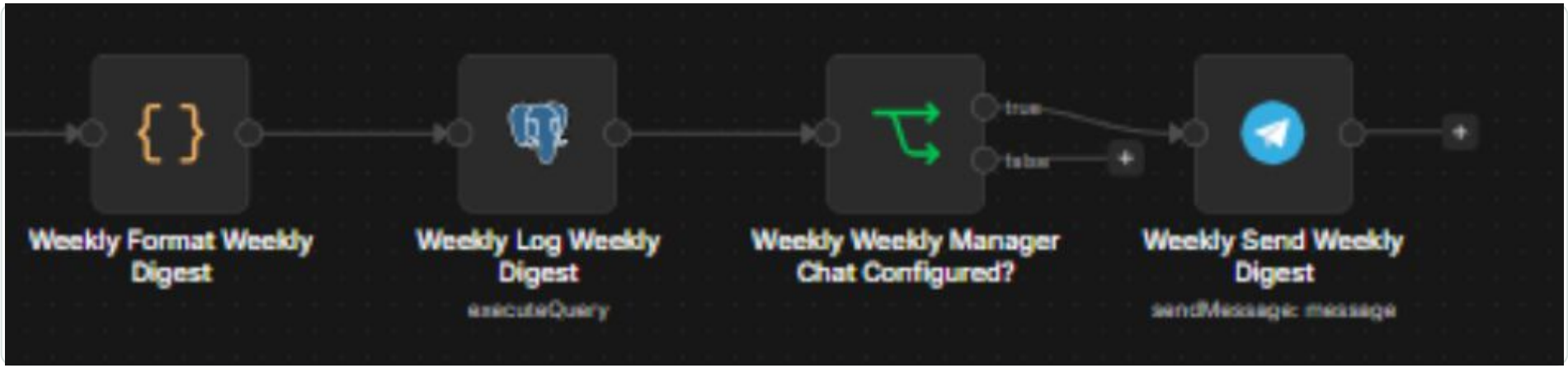
The lower lane reports observed activity using the same stored metrics.

WEEKLY / LOAD METRICS



A scheduled run loads vendor reply and handling metrics.

WEEKLY / FORMAT AND SEND

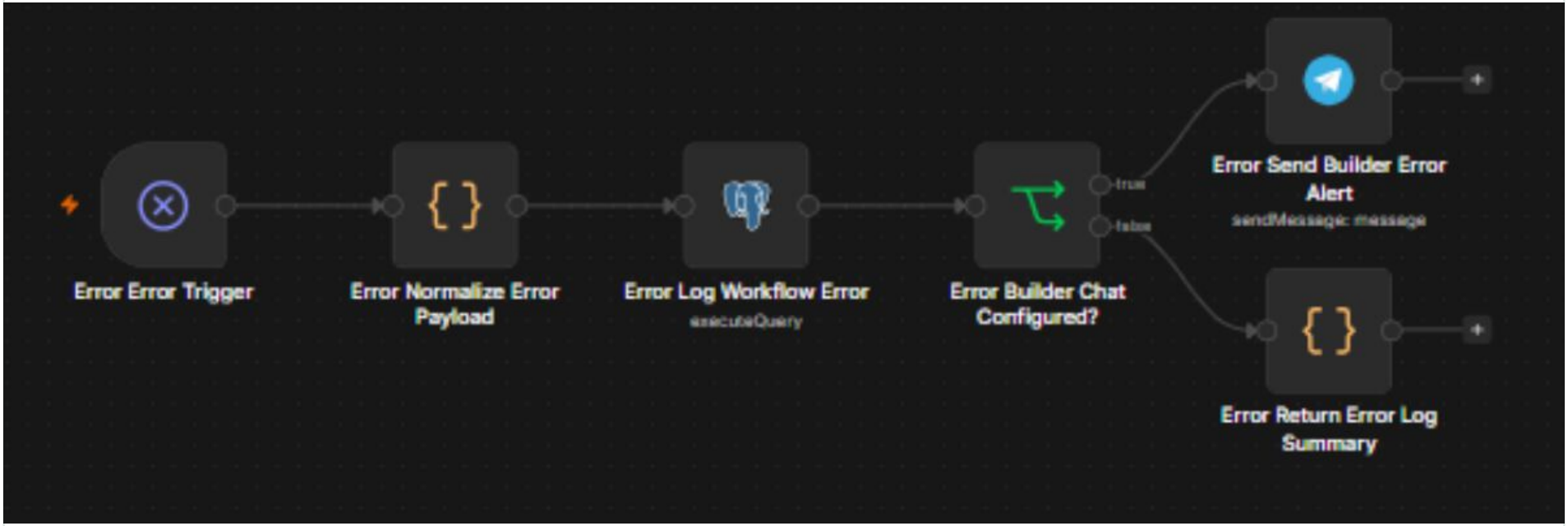


The message is logged before the manager receives the summary.

WORKFLOW / BUILDER-ERROR-HANDLER / 6 NODES

Builder Error Handler

Unexpected workflow failures are recorded separately from normal manager alerts.



TRIGGER

An n8n workflow error.

KEY STEPS

Normalize the error and write an event row.

DECISION

Check whether the private builder alert chat is configured.

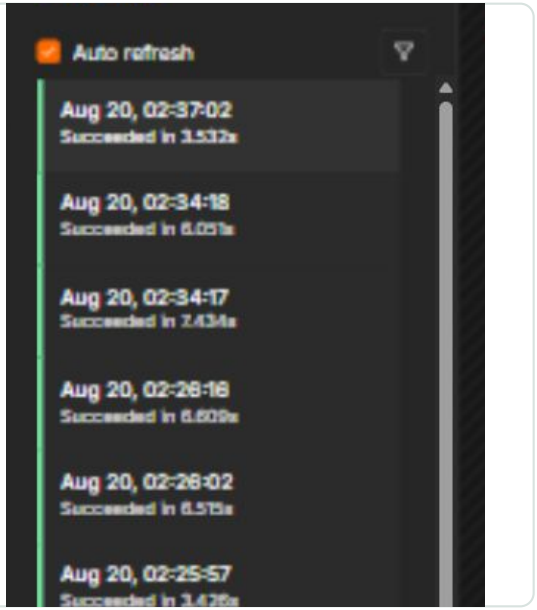
OUTPUT

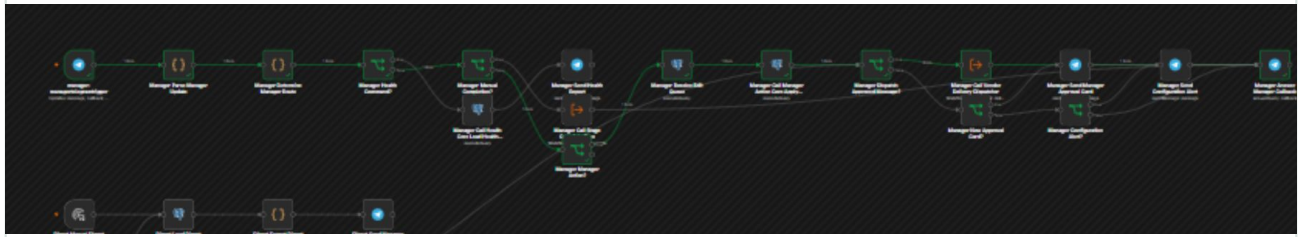
A concise builder alert and stored error summary.

VERIFICATION

Tested with synthetic scenarios

The evidence below is fixture testing and live local executions, not a customer outcome.





LIVE N8N EXECUTIONS / SUCCESS STATUS AND RUN PATH

20 / 20

The guided F01-F20 run passed across Telegram, generic inbound and manager controls.

16 / 16

Separate email and message-interpretation scenarios passed.

WHAT WAS OBSERVED

- An ambiguous unit message asked the manager; it did not advance a stage.
- A duplicate vendor completion produced one stage change and one approval draft.
- A Deep Clean report was blocked while Repairs remained open.
- A manager-approved single channel created one delivery row and activated the next stage.
- A full mirror check matched six synthetic turns and 66 stages with linked records populated.